

Implementation of the KNN Algorithm for Food Recommendation System Based on Users' Nutritional Needs

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Article history:

Received 04 Dec 2025;
Revised 08 Dec 2025;
Accepted 10 Dec 2025;
Available online 10 Dec 2025

Keywords:

BMR
Food Recommendation System
K-Nearest Neighbors (KNN)
Nutritional Needs
TDEE

Abstract

This study develops a web-based food recommendation system using the K-Nearest Neighbors (KNN) algorithm to provide personalized food recommendations based on users' nutritional needs and preferences. Many individuals struggle to create balanced diets due to insufficient knowledge or time, which can lead to malnutrition or obesity. To address this, the system calculates users' nutritional needs using Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE), incorporating preference filtering provided by users. The KNN algorithm then analyzes a food database to identify items that best match the users' nutritional profiles. The system features two primary interfaces: a user interface for inputting nutritional data and displaying recommendations, and an administrative interface for managing food data, user information, and recommendation history. The system was evaluated through Black Box Testing, which confirmed that all main features function as intended. The KNN algorithm demonstrated effectiveness by providing relevant food recommendations that align with users' individual nutritional requirements. Key evaluation metrics, such as recommendation accuracy and user satisfaction, validate the system's performance. This approach highlights the system's potential in offering personalized nutrition advice, with a focus on real-time decision-making. Future work will aim to incorporate additional dietary factors and expand the food database to enhance the system's adaptability and precision.

I. INTRODUCTION

Health and nutritional balance are key factors in supporting the quality of life of the community, especially in the modern era where awareness of the importance of a healthy diet continues to increase. However, a significant challenge remains: achieving nutritional balance requires an understanding of individual needs, which is often lacking due to limited knowledge and time to plan an ideal diet. The Government of Indonesia, through the Ministry of Health, has implemented various health development programs, one of which is the Free Lunch Program for school students. This program is designed as a strategic effort to reduce the prevalence of malnutrition and improve nutritional adequacy in children from an early age, with the expectation of breaking the chain of nutritional problems such as stunting and wasting, which are still prevalent in Indonesia [1]. These challenges underscore the necessity for practical, accessible solutions that can guide individuals in making healthier food choices.

According to a report by the World Health Organization (WHO), global nutrition problems remain a major challenge, with more than 2.5 billion adults being overweight and 390 million adults being underweight, while among children under five years of age, there are 149 million cases of stunting, 45 million children suffering from wasting, and another 37 million experiencing overweight or obesity [2]. This reflects the complexity of nutritional problems, which requires a deeper understanding and an integrated approach, including the utilization of information technology as an efficient medium for education and intervention. Digital health literacy has shown

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to positively influence individuals' ability to meet balanced nutrition needs, making digital platforms a valuable tool for nutritional education and behavior change [3].

Various community service and research studies have shown that structured balanced nutrition education can improve public knowledge and attitudes toward the prevention of obesity, stunting, and the double burden of malnutrition among adolescents and families [4][5]. Educational activities that integrate balanced nutrition materials, reading nutritional value information labels, and utilizing digital media have been proven to enhance participants' ability to understand nutrient composition and its implications for health [6]. While these efforts are valuable, they often fail to provide real-time, personalized support for individuals in their daily food choices.

To address this gap, this study focuses on the development of a web-based food recommendation system that integrates real-time nutritional calculations, such as Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE), to help users monitor their daily energy needs. The system will also allow for the dynamic adjustment of recommendations based on users' individual preferences, goals, and activity levels. However, many existing implementations still focus on presenting information and classifying nutritional status; therefore, further development is needed toward systems capable of providing more personalized menu recommendations by adjusting to users' preferences and goals.

The main obstacle in achieving nutritional balance is the lack of individual understanding of their daily caloric needs. The body's energy requirements can be calculated based on Basal Metabolic Rate (BMR) and the level of physical activity. BMR describes the minimum amount of energy required by the body to perform basic functions, while the level of physical activity determines the additional energy needed for daily activities. An imbalance between the energy consumed and the energy required by the body can lead to serious health consequences [7]. Excessive calorie consumption can increase the risk of obesity, which in turn can lead to various chronic diseases such as type 2 diabetes, hypertension, and heart disease. Conversely, prolonged calorie deficits can cause health disorders such as anemia, osteoporosis, and decreased immune function.

In line with the increasing variety of food types, ranging from fast food to organic food, individuals are increasingly faced with the challenge of choosing foods that match their nutritional needs. Therefore, an intelligent system is needed that can help users understand their energy requirements and provide food recommendations that are tailored to everyone's nutritional needs and preferences. This study aims to bridge that gap by developing a personalized food recommendation system using the K-Nearest Neighbors (KNN) algorithm to suggest foods that align with users' nutritional profiles.

This study aims to develop a web-based food recommendation system that uses the K-Nearest Neighbors (KNN) algorithm and a content-based filtering approach to provide food suggestions that match users' nutritional needs and preferences. The system calculates nutritional requirements based on Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE), then applies the KNN algorithm to classify individuals according to their calorie needs and match them with the most suitable food data. Through this approach, the system is expected to provide relevant and accurate recommendations.

The system developed in this study consists of two main components, namely the user side and the admin side. On the user side, the system provides an interface that allows users to input personal data such as age, gender, weight, height, activity level, and food preferences. Based on these data, the system will provide food recommendations that match everyone's nutritional needs. Meanwhile, on the admin side, the system provides features to monitor food data, manage user information, and view the history of recommendations received by users. By incorporating these functionalities, this study aims to contribute to the development of personalized, real-time nutrition decision support tools using AI-based algorithms.

Through the development of this recommendation system, it is expected that people will find it easier to understand eating patterns that are appropriate to their physical condition and make better decisions when choosing food. In addition, this study is expected to contribute to the development of artificial intelligence-based recommendation systems in the fields of health and nutrition, as well as serve as a reference for the application of information technology to improve people's quality of life.

II. RELATED WORKS/LITERATURE REVIEW

One study developed a system that calculates energy requirements using BMR and TDEE, then applies the KNN algorithm to group users according to their nutritional profiles, enabling the system to match individuals with foods that align with their nutritional composition and health constraints such as allergies and diabetes. The evaluation results, which report an RMSE of 18.75, indicate that KNN is capable of accommodating variations in individual nutritional needs within food recommendation scenarios [8]. However, the system's focus is on general nutritional matching rather than personalization based on dynamic user preferences, limiting its scalability.

Another study focuses on Indonesian traditional foods as the main recommendation targets, employing KNN to identify items that closely correspond to users' nutritional requirements such as specific protein, fat, vitamin, and mineral levels. The findings highlight KNN's potential to support initiatives such as stunting prevention, although the system remains relatively simple and has not yet been integrated into a large-scale web-based platform [9]. While this approach emphasizes culturally relevant food choices, it lacks the ability to offer

personalized recommendations based on individual metabolic needs or goals, which limits its practical application.

A different study applies the KNN algorithm to detect eating patterns that may pose health risks within specific populations. By processing daily nutritional intake data, the system classifies individuals into healthy or high-risk categories; however, the focus remains on risk identification rather than on providing personalized dietary recommendations tailored to individual nutritional needs [10]. This system's primary limitation is its focus on risk assessment rather than actively supporting users in achieving personalized health outcomes, which this study aims to improve.

Another study develops a food recommendation application specifically for pregnant women, using KNN to tailor suggestions according to gestational age, trimester-specific energy requirements, and maternal health conditions. The application improves adherence to recommended dietary patterns, yet its scope remains limited due to its Android-only implementation and narrow target population [11]. While useful in specific contexts, its narrow focus and platform limitation reduce its scalability and broader applicability to other populations.

A further study aims to categorize foods based on nutritional similarity by integrating clustering techniques with KNN. This approach produces nutritionally coherent food groups that can be used as a foundation for balanced menu planning. However, the system does not yet provide personalized recommendations that incorporate individual BMR, TDEE, or detailed user preference profiles [12]. This limitation underscores the need for more tailored systems that incorporate individual energy requirements and preferences, which is central to the current study's development.

Across these studies, it is evident that KNN has been widely and effectively applied in nutrition-related contexts ranging from nutritional status classification and food clustering to recipe and menu recommendation systems. However, despite these advances, most existing works still lack robust mechanisms for personalizing recommendations based on real-time BMR and TDEE calculations within a fully implemented web-based environment. Many systems also target specific populations or operate on limited testing scales, reducing their potential for broader adoption across diverse user groups with varied dietary behaviors. This gap in personalization and scalability is the primary focus of this study.

Furthermore, previous research often prioritizes technical performance indicators such as accuracy, precision, or prediction error while paying less attention to user experience, cultural food preferences, or contextual relevance in real-world dietary decision-making. Moreover, systems rarely integrate feedback loops or longitudinal monitoring of eating behaviors, which are crucial for ensuring that recommendations effectively support healthier, sustainable dietary habits. For these reasons, the present study titled "Implementation of the K-Nearest Neighbors Algorithm in Food Recommendation Based on Nutritional Needs" seeks to address these gaps by developing a web-based system that incorporates individualized energy requirement calculations and KNN to produce more personalized, context-aware, and practically applicable food recommendations for daily use.

III. METHODS

This study uses a software engineering approach by developing a web-based food recommendation system using the Flask framework and a MySQL database. The system is designed to provide food recommendations based on energy requirement calculations (BMR and TDEE), user goals (dieting or gaining weight), physical activity, dietary restrictions, and preferences (vegetarian, non-vegetarian, or unrestricted). The research method is structured chronologically, starting from requirements analysis, system design, implementation of the K-Nearest Neighbors (KNN) algorithm, through to functional testing of the recommendation system.

A. Data dan Basis Data

The primary data, consisting of food items and their macronutrient values, were obtained from an open GitHub repository (NuSa-Nutrition-Scan/Food-Recommendation), then downloaded in CSV format and further processed. The main dataset contains the food name, source, energy, protein, fat, and carbohydrates. The data were cleaned, enriched with a vegetarian category attribute, and then imported into a MySQL database in a food table. The primary data consist of a list of foods along with their energy, protein, fat, and carbohydrate contents, which are then inserted into the food table in the database. Web-based food menu recommendation systems are widely used to help users select menus that match their preferences and needs. The food attribute variables used in this study are presented in Table 1.

TABLE 1
RESEARCH VARIABLES

Variables	Description	Type Of Data
Food	Food Name	Varchar
Source	Food Ingredients	Varchar
Energy	Energy/Calories in Food	Float
Protein	Protein Content	Float
Fat	Fat Content	Float

Carbohydrates Vegetarian	Carbohydrate Content Food Preferences	Float Tinyint
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The database also includes the users, hasil_perhitungan, and hasil_rekomendasi_baru tables to store accounts, nutritional profile inputs, and recommendation results.

B. Nutritional Needs Analysis

Nutritional needs represent the minimum amount of nutrients required by an individual to maintain optimal body function. These nutritional intakes primarily come from food consumed daily, where the balance between intake and nutritional requirements greatly influences a person's nutritional status. The main component of nutritional needs is macronutrients or energy requirements, which can be calculated using Basal Metabolic Rate (BMR) as the primary indicator. BMR describes the minimum amount of energy the body needs to perform vital functions at rest, and its value is influenced by factors such as age, gender, weight, and height [13]. The BMR calculation in this study uses the Harris-Benedict formula as described in equations (1) and (2).

For Men:

$$BMR = 66 + (13,7 * W) + (5 * H) - (6,8 * A) \quad (1)$$

For Women:

$$BMR = 655 + (9,6 * W) + (1,8 * H) - (4,7 * A) \quad (2)$$

Description:

W = Weight (Kg)

H = Height (Cm)

A = Age (Years)

After obtaining the Basal Metabolic Rate (BMR) value, the next step is to calculate the Total Daily Energy Expenditure (TDEE) or total daily energy requirement. While BMR represents the minimum energy needed by the body to maintain vital functions at rest, TDEE includes all the energy required for daily activities and exercise. The TDEE value is calculated by multiplying BMR by a physical activity factor (activity multiplier), using the Katch-McArdle formula approach [14]. It can be seen in the following Table 2.

TABLE 2
LEVEL OF PHYSICAL ACTIVITY

Activity Level	Formula
Sedentary	BMR x 1.2 (Sitting all day at the office or at home, driving, and only walking lightly around the house or office. Not exercising at all or only once a month)
Slightly Active	BMR x 1.375 (Engage in light exercise 1–3 times a week for 30 to 60 minutes. Examples include brisk walking, casual cycling, or light yoga. Daily activities involve a lot of standing and walking.)
Moderately Active	BMR x 1.55 (Engaging in moderate exercise 6–7 times a week for more than 60 minutes. Examples include running, swimming, aerobics, soccer, and badminton. Daily activities involve a lot of moving around and standing.)
Very Active	BMR x 1.725 (Doing heavy exercise every day or twice a day. Examples include long-distance running, weightlifting, intensive aerobics, and team sports. Jobs that require a lot of movement and heavy physical activity.)
Extra Active	BMR x 1.9 (Engaging in intense exercise twice or more in a day. For example, professional athletes, or heavy physical laborers like construction workers. Almost all of the time is spent on intense physical activity.)

The body requires nutrient intake to support its various biological functions. Nutrients are divided into two main groups, namely macronutrients (protein, carbohydrates, and fats) and micronutrients (vitamins and minerals). In this study, the analytical focus is directed toward macronutrient requirements. Everyone's nutritional needs differ depending on factors such as age, sex, body weight, height, level of physical activity, and health status. The calculation of daily nutritional requirements in this study refers to the formulas established by the Institute of Medicine (IOM), which can be seen in the following equations.

$$Carbohydrates = \frac{65\% * Daily \text{ calorie needs}}{4} \quad (3)$$

$$Proteins = \frac{15\% * Daily\ calorie\ needs}{4} \quad (4)$$

$$Fats = \frac{20\% * Daily\ calorie\ needs}{9} \quad (5)$$

C. K-Nearest Neighbors Algorithm in Recommendation Systems

A recommendation system is a system that predicts items that are relevant to users based on patterns of preferences or item characteristics. In this study, a content-based approach is used with the K-Nearest Neighbors (KNN) algorithm, which works by finding several nearest neighbors (k) based on the similarity of nutritional features between users' nutritional requirements and the nutritional profiles of foods [15][16]. A food recommendation system is an information technology application designed to help users select types of food that are appropriate to their preferences, nutritional needs, health conditions, or specific dietary restrictions. This system operates by processing user data such as age, body weight, height, level of physical activity, and medical history, then matching them with the available food data to generate the most relevant recommendations [13].

The main objective of the K-Nearest Neighbor (KNN) algorithm is to identify a new object by considering its closest distance to several previously existing data neighbors. In the context of this study, the objects being classified are the nutritional contents of foods, which include calories, protein, fat, and fiber. The proximity between data points is measured using the Euclidean distance method. The equation used to calculate this distance is presented in Equation 6 below.

$$d(x, y) = \sqrt{\sum_{k=1}^n (x_k - y_k)^2} \quad (6)$$

Euclidean distance is a method used to measure the degree of similarity or closeness between two data points [17]. The Euclidean distance formula is shown in Equation (6), where D represents the distance between two points, namely a point in the training data (x) and a point in the test data (y) to be classified. The vectors x and y each consist of attribute elements, namely $x_1, x_2, \dots, x_i$ and $y_1, y_2, \dots, y_i$, with k as the number of dimensions or attributes being compared. As an example of its application, if the distance between two food items such as chicken satay and meatballs is to be calculated, the attributes used may include calories, protein, carbohydrates, and fat. In this case, x_1 to x_4 represent the attribute values for chicken satay, and y_1 to y_4 represent the attribute values for meatballs. The distance between the two is calculated using the Euclidean distance formula in the following equation.

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + (x_3 - y_3)^2 + (x_4 - y_4)^2} \quad (7)$$

D. Food Recommendations for Users

This study categorizes users into three main conditions in the food recommendation process. First, users in a healthy condition, for whom the system generates food recommendations that match their energy and nutrient needs based on the calculated BMR and TDEE, without any specific restrictions. This type of personalization is in line with recent web- and mobile-based personalized nutrition applications, which deliver automated dietary advice tailored to individual profiles rather than relying on generic guidelines [18].

Second, users with certain food allergies or dietary restrictions. In this category, the system automatically filters the list of foods by excluding ingredients or food sources listed in the user's restriction list, so that the recommendations provided are safe for consumption. This rule-based handling of allergies and preferences is consistent with contemporary nutrition recommender systems, which explicitly remove disallowed items from the candidate set to maintain both safety and usability of the recommendations [19].

Third, users with special preferences or needs, such as vegetarians, non-vegetarians, or users with specific goals such as dieting or gaining weight. The system adjusts the total daily calorie intake and macronutrient composition (carbohydrates, protein, and fat) according to these goals and preferences, so that the recommended foods are more aligned with users' metabolic needs and lifestyle choices. This approach reflects current developments in personalized food recommendation research, where systems integrate health status, dietary preferences, and dietary taboos to support individualized wellbeing-oriented food choices [20].

IV. RESULTS

A. Implementation

In this study, we implement a web-based food recommendation system built using the Flask framework with a MySQL backend, employing the K-Nearest Neighbors (KNN) algorithm to generate dietary suggestions tailored to users' nutritional needs, goals, activity levels, dietary restrictions, and preferences (vegetarian vs. non-vegetarian). The implementation proceeds through several stages: integrating nutritional requirement calculations (BMR and TDEE), filtering food data, normalizing nutritional features, and invoking the KNN model to produce

a ranking of recommended foods. This approach is consistent with recent trends in personalized nutrition systems that combine nutrient-based filtering and machine learning to deliver individualized meal suggestions [21].

Operationally, the implementation flow begins with user registration and login, after which users complete a nutrition profile form on the dashboard page. On this form, they input anthropometric data (body weight, height, age, sex), physical activity level, dietary goal (weight loss, maintenance, or weight gain), a list of food restrictions, and their vegetarian or non-vegetarian preference. Once submitted, the system calculates the Basal Metabolic Rate (BMR) using a standard equation, computes TDEE by applying an activity factor, and then adjusts TDEE according to the selected goal. From these calculations, an energy and macronutrient requirement vector (calories, protein, fat, carbohydrates) is constructed as the query point for the KNN procedure. Similar pipelines that derive calorie needs from BMI/BMR and TDEE before generating diet plans have been shown to be effective in guiding individualized meal recommendations [22].

The food data used by the system are sourced from an open dataset (hosted on GitHub), preprocessed, and stored in a MySQL food table. Each record contains the food name, source, energy, protein, fat, and carbohydrate content, along with a vegetarian flag indicating whether the item belongs to the vegetarian category. Prior to similarity computation, the system applies a filtering layer that removes foods incompatible with the user's restriction list (e.g., allergenic or disallowed ingredients) and enforces the user's vegetarian, non-vegetarian, or unrestricted preference. Only items that satisfy these constraints are passed to the subsequent numerical processing stage. This combination of structured nutrition data, feature preprocessing, and constraint-based filtering follows current best practices in nutrition-aware recommender systems, where standardized nutrient features are used to improve similarity calculations and ensure that recommendations comply with user-specific constraints [23].

At the modeling stage, the system constructs a feature matrix from the nutritional attributes (energy, protein, fat, carbohydrates) and normalizes all features using standardization so that each attribute lies on a comparable scale. The user's nutritional requirement vector derived from BMR and TDEE is normalized with the same parameters to maintain consistency. The KNN algorithm is then applied with a fixed neighborhood size of $k = 5$, using Euclidean distance to measure similarity between the user's normalized requirement vector and the normalized nutritional profiles of candidate foods. The five food items with the smallest distances are selected as the most relevant recommendations. The choice of KNN with distance-based ranking aligns with recent work in food recommender systems that employ KNN (alone or in hybrid architectures) to identify nutritionally similar items and provide personalized food suggestions [24].

the implementation includes a structured logging mechanism to support traceability and future evaluation. The table `hasil_perhitungan` stores the username, anthropometric data, physical activity level, goal, list of restrictions, preferences, and the resulting BMR and TDEE values. The recommended food list for each computation session is stored in the `hasil_rekomendasi_baru` table, linked via `perhitungan_id`. Thus, every time a user performs a calculation, the system both displays the recommendations in the web interface and records the associated inputs and outputs for longitudinal monitoring. Such persistent storage of user profiles, energy requirements, and recommended meals is in line with contemporary personalized nutrition frameworks that emphasize historical tracking to enable ongoing assessment and refinement of diet recommendation strategies [25].

The main user page begins with a login and registration form, which allows users to create a new account and access the recommendation dashboard. After successfully logging in, users are directed to the dashboard page that displays a nutrition profile input form containing body weight, height, age, sex, physical activity level, goal (weight loss, maintenance, or weight gain), food restrictions, and vegetarian preference. After the user clicks the process button, the system displays the recommendation results page, which includes the BMR value, TDEE, and a list of five recommended foods along with information on energy, protein, fat, and carbohydrates. The page layout can be seen in Figure 1 below.

(a) Login Page

(c) Home Page



Fig. 1 User Interface

The system provides a history page that allows users to review previous calculation sessions. On this page, each history entry displays the calculation time, the user profile data used, the BMR and TDEE values, and the list of foods recommended in that session. The history is arranged chronologically based on access time, enabling users to monitor changes in recommendation patterns in line with changes in body weight, physical activity, or dietary goals. The page layout can be seen in Figure 2 below.

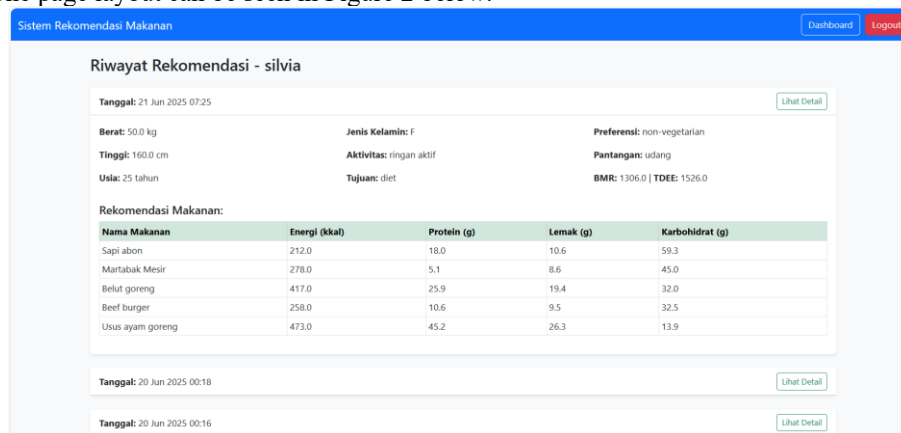
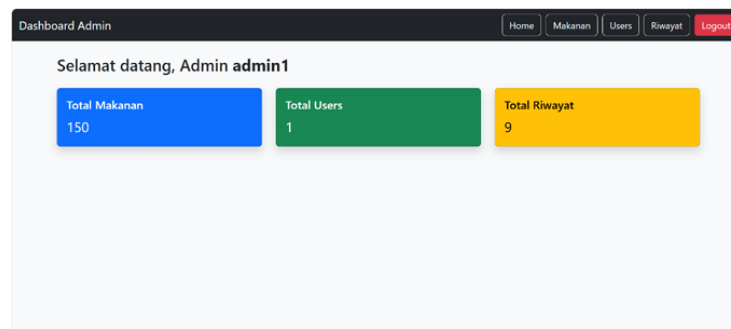


Fig. 2 User History Display

From the administrator's perspective, the system provides an admin dashboard page that displays a summary of the number of food data entries, the number of users with the user role, and the number of stored calculation histories. The admin can access the food list page to view, add, or update food data that will be used in the recommendation process. In addition, there is a user list page to monitor registered accounts, and a global recommendation history page that displays all calculation results and the lists of recommendations generated by the system for all users. The admin page layout can be seen in Figure 3 below.



(a) Admin Dashboard Page

Admin Panel						
Daftar Makanan						
ID	Nama	Energi (kcal)	Protein (g)	Lemak (g)	Karbohidrat (g)	Vegetarian
1	Bakso	202.0	12.41	13.16	7.58	Tidak
2	Bubur Ayam	155.0	11.48	5.16	15.05	Tidak
3	Mi Goreng	475.0	10.0	15.0	65.0	Ya
4	Sate	225.0	19.54	14.82	4.87	Tidak
5	Soto	130.0	9.96	6.19	8.11	Tidak
6	Sop buntut	71.0	7.5	3.6	2.2	Tidak
7	Sop daging sapi	49.0	5.8	0.5	5.3	Tidak
8	Sop kaki sapi	68.0	9.0	1.9	3.7	Tidak
9	Sop kambing	34.0	5.5	0.8	1.1	Tidak
10	Sop konro	71.0	7.4	2.6	4.5	Tidak

(b) Food List Page

Admin Panel			
Daftar Pengguna			
ID	Username	Email	Role
1	silvia	silvia12@gmail.com	user
2	admin1	admin1@gmail.com	admin

(c) User List Page

Admin Panel

Home

Makanan

Users

Riwayat

Login

Riwayat Rekomendasi - Semua Pengguna

Username: Silvia | Tanggal: 19 Jun 2025 07:25

Detail

Berat: 50.0 kg

Tinggi: 160.0 cm

Usia: 25 tahun

Jenis Kelamin: F

Aktivitas: ringan aktif

Tujuan: diet

Preferensi: non-vegetarian

Pantangan: udang

BMR: 1306.0 | TDEE: 1526.0

Rekomendasi Makanan:

Nama Makanan	Energi	Protein	Lemak	Karbohidrat
Sapi abon	212.0 kcal	18.0 g	10.6 g	59.3 g
Martabak Meja	278.0 kcal	5.1 g	8.6 g	45.0 g
Selut goreng	417.0 kcal	25.9 g	19.4 g	32.0 g
Beef burger	258.0 kcal	10.6 g	9.3 g	32.5 g
Urus ayam goreng	473.0 kcal	45.2 g	26.3 g	13.9 g

Username: Silvia | Tanggal: 20 Jun 2025 00:18

Detail

Username: Silvia | Tanggal: 20 Jun 2025 00:16

Detail

(d) Recommendation History Page
Fig. 3 Admin Interface

B. System Testing

System testing in this study uses the black box testing method. Black box testing is a type of software testing that focuses on examining system functions based on the inputs and the outputs produced, without reviewing the internal structure of the program or the details of its code implementation [26]. This approach is widely used in the development of web-based applications and information systems because it can validate whether each feature operates in accordance with the functional requirements defined in the design. This can be seen in Table 3.

TABLE 3
BLACKBOX TESTING

Test Description	Test Scenario	Expected Results	Results Obtained	Tester
Registration	Filling in all fields (username, email, password)	A new account is saved in the database and the user is redirected to the login page	Appropriate	User
Login	Leaving one field empty (for example, email)	The system rejects the save and displays an error message	Appropriate	User
	Entering the correct email and password	Displaying the admin homepage	Appropriate	Admin
	Entering the wrong email and password	Login failed	Appropriate	Admin

Logout	Entering the correct email and password	Displaying the user homepage	Appropriate	User
	Entering the wrong email and password	Login failed	Appropriate	User
	Press the logout button	Logout of the system and display the login page	Appropriate	Admin
	Press the logout button	Log out of the system and display the login page	Appropriate	User
Input Nutrition Profile	All fields are filled (weight, height, age, etc.)	The system saves input, calculates BMR and TDEE, and redirects to the recommendation page	Appropriate	User
	Leaving one field empty (for example, weight)	The system rejects the process and displays an error message	Appropriate	User
	Filling in food restrictions	The system removes foods containing restricted keywords from the candidate recommendation list	Appropriate	User
	Selecting vegetarian preference	Displays vegetarian food	Appropriate	User
Recommendation Process	Selecting non-vegetarian preference	Displays non-vegetarian food	Appropriate	User
	Valid input and sufficient food data	The system displays 5 food recommendations along with BMR and TDEE values, and saves the data to history	Appropriate	User
	Accessing the history page after previously receiving recommendations	The system displays a history list containing calculation time, BMR, TDEE, and a list of recommended foods	Appropriate	User
	Accessing the history page without ever performing calculations	The system displays information that history is not available or the history list is empty	Appropriate	User
Manage User and Admin Data	Changing data by filling in all user data fields	Data is stored in the database	Appropriate	Admin
	Changing, adding, and deleting data in the food list and Recommendation History columns	Data is successfully modified, added, and deleted from the food list and Recommendation History	Appropriate	Admin

V. DISCUSSION

The implementation results demonstrate that the developed food recommendation system successfully integrates nutritional requirement calculations—namely BMR and TDEE—together with physical activity levels, dietary goals, food restrictions, and vegetarian or non-vegetarian preferences into a unified recommendation workflow based on the K-Nearest Neighbors (KNN) algorithm. The system consistently produces food lists whose nutritional profiles closely match users' energy and macronutrient needs, while also adhering to the specified filtering constraints. This outcome aligns with previous findings showing that KNN is capable of matching user nutritional requirements and generating relevant food suggestions when applied within a structured nutritional feature space [8].

Black box testing of the system's main functionalities, including registration, login, nutrition profile input, recommendation generation, and history management, further shows that each feature operates according to the design specifications. Valid inputs consistently yield correct outputs, while invalid or incomplete inputs are appropriately handled through system validation mechanisms. These results highlight that both the user interface and backend logic effectively support the system's objective of offering a practical web-based tool that facilitates informed food selection based on personalized nutritional profiles. Such functional robustness is consistent with prior work in diet-related classification tasks, where KNN has proven reliable in processing nutritional data and differentiating health-related patterns with reasonable accuracy [10].

When compared with earlier studies employing KNN in nutrition domains, this research distinguishes itself through its integration of direct energy requirement calculations (BMR and TDEE) before applying KNN to the nutritional feature matrix. Many previous studies either concentrated solely on nutrient similarity or focused on clustering foods without personalizing them to user-specific metabolic needs. For instance, research combining clustering methods with KNN has shown that nutritional similarity can successfully generate coherent food groups for menu planning, though without personalized recommendation capabilities linked to individual energy needs [12]. In contrast, the current study incorporates personal caloric targets, dietary goals, and preference filtering prior to similarity computation, allowing the recommendation process to be more aligned with real-world dietary decision-making.

Despite these strengths, several limitations remain. The food dataset used in this study is limited to macronutrients and does not include micronutrient information or clinical dietary considerations, thereby constraining the system's ability to support users with specific medical conditions. The preference and restriction filtering mechanisms also rely on simple keyword matching and a binary vegetarian attribute, which restricts the system from handling more complex dietary patterns or detailed allergen tracking. These limitations could impact the system's real-world applicability, particularly for users with more nuanced dietary needs or health conditions. In addition, the KNN implementation uses a fixed k value without examining the potential impact of varying k or adopting alternative distance metrics on recommendation quality.

Future development may therefore consider expanding the food database to incorporate micronutrients and disease-specific dietary guidelines, enabling more comprehensive nutritional recommendations. Preference modeling can be improved by integrating user feedback loops or historical consumption patterns, allowing the system to adapt over time to better align with individual dietary habits. Over time, user preferences may evolve, and a more dynamic adaptation system would enhance long-term utility. From the algorithmic perspective, further studies may explore optimization of the k parameter, evaluate different similarity measures, or combine KNN with hybrid recommendation methods to enhance accuracy and personalization. These enhancements would strengthen the system's alignment with broader applications in personalized nutrition and data-driven dietary support. The implications for public health are significant, particularly in promoting healthy eating habits and combating issues like malnutrition and obesity. In educational settings, such a system could serve as a practical tool for nutritionists, educators, and health professionals to provide real-time, tailored dietary guidance to individuals, improving overall health literacy.

VI. CONCLUSIONS

Based on the results of the research that has been carried out, the web-based food recommendation system developed in this study successfully applies the K-Nearest Neighbors (KNN) algorithm to provide food recommendations according to individual nutritional needs. The system is able to calculate users' daily energy requirements through BMR and TDEE, then integrate them with physical activity level, dietary goals (weight loss, maintenance, or weight gain), food restrictions, and vegetarian or non-vegetarian preferences into a single, unified recommendation flow. This integration of BMR and TDEE with a dynamic recommendation engine represents the novelty of the system, enabling personalized, real-time dietary guidance based on each user's specific nutritional profile.

The tests show that all main functions, including user account management, nutrition profile data input, calculation and recommendation processes, as well as history display, operate in accordance with the design. The validation features also work properly, so that incomplete or invalid inputs can be handled appropriately by the system. Based on these achievements, this study provides a solid foundation for further development in the field of nutrition recommendation systems. In the future, the system can be refined by expanding the coverage of food data, incorporating micronutrient parameters, specific health conditions, and integrating wearable health devices to provide more precise real-time recommendations. Additionally, incorporating hybrid algorithms could enhance recommendation accuracy by combining KNN with other techniques such as collaborative filtering or deep learning models. With such further developments, the system is expected to provide recommendations that are increasingly precise, scalable, and adaptable to user needs in various healthcare, dietary, and mobile app contexts.

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